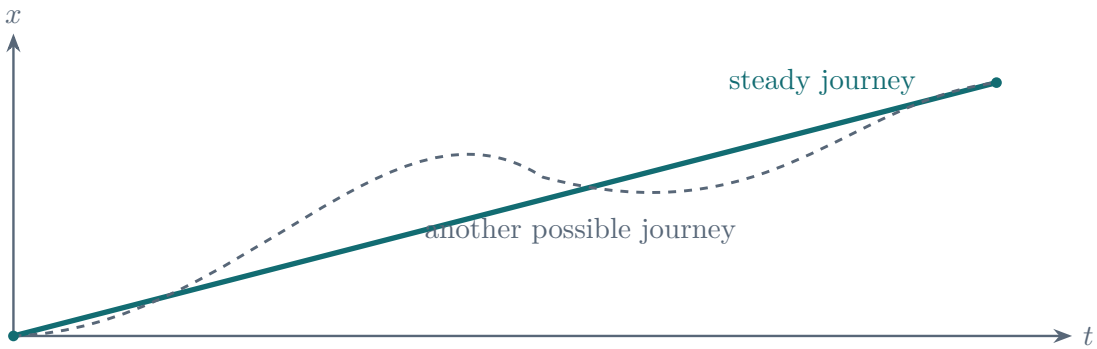
Why $T - V$?

A gentle route to the action and Lagrangian mechanics

From a free particle, to work, to a balance between two effects.



A conceptual explanation, not a problem-solving manual.

We will distinguish what the intuition suggests from what the principle actually states.

1. Begin with a free particle

Imagine a particle travelling from A to B in a fixed time τ. No forces act on it.

We know what happens: it moves at constant velocity. But instead of starting with Newton's first law, let us ask a different question:

What is mathematically special about the steady journey?

There are many imaginable journeys with the same starting point, ending point, and duration. The particle could rush early and slow later. It could begin slowly and catch up. All would arrive on time.

A very small numerical example

Suppose the distance is 10 m and the time is 2 s. The average velocity must be 5 m/s.

Journey	First second	Second second	Average v^2
Steady	5	5	25
Uneven	3	7	29

Both journeys cover 10 m. Yet the uneven one has a larger average of v^2.

Why does that matter? Because kinetic energy is

$$T = \frac{1}{2}mv^2.$$

The square makes velocity fluctuations expensive in a precise mathematical sense: a period of extra speed is not fully compensated by an equally long period of reduced speed.

First intuition. Among journeys with the same endpoints and duration, steady motion has the smallest *accumulated kinetic energy*, $\int T \, \mathrm{d}t$.

2. What does "unnecessary variation" mean?

Let the average velocity be $\bar{v} = (B - A)/\tau$. We can describe any other velocity profile as

$$v(t) = \bar{v} + u(t),$$

where $u(t)$ is the deviation from the average. Because the total displacement is fixed, $\int_0^\tau u(t) \, \mathrm{d}t = 0$.

Then

$$\begin{aligned}\int_0^\tau T\,\mathrm{d}t &= \frac{m}{2}\int_0^\tau (\bar{v}+u)^2\,\mathrm{d}t \\ &= \underbrace{\frac{m\tau}{2}\bar{v}^2}_{\text{fixed by the journey}} + \underbrace{\frac{m}{2}\int_0^\tau u(t)^2\,\mathrm{d}t}_{\text{extra contribution from fluctuations}} .\end{aligned}$$

The second term cannot be negative. It vanishes only for $u(t) = 0$, or constant velocity.

This is the free-particle result in one line of reasoning. We have not said that the particle is trying to save fuel or reduce its instantaneous energy. We have simply discovered a quantity whose minimum describes its motion.

3. Change the question: where can a particle rest?

Now place a particle in a force field. Before allowing it to move, ask where it could remain at rest.

Think of a ball at the bottom of a smooth bowl. If we displace it slightly, it tends to return. Compare that with a ball delicately balanced on top of a hill: it can be at rest, but a slight displacement sends it away.



bottom of bowl
stable equilibrium

top of hill
unstable equilibrium

At either exact equilibrium point, the net force is zero:

$$F(x_0) = 0.$$

How can we express this using *work*?

For a tiny displacement δx, the work done *by the force* is approximately

$$\delta W = F(x_0)\,\delta x.$$

So, at equilibrium, there is no *first-order* work for a small displacement.

> **Second intuition.** A particle can remain at rest where a tiny displacement produces no first-order work. But this alone means *equilibrium*, not necessarily *stability*. At stable equilibrium, a small displacement requires positive work *against* the force.

4. Exactly what do we mean by W?

This distinction is essential for everything that follows.

For a *conservative* force, choose a reference position x_r. Define

$$\boxed{W(x) = \int_{x_\mathrm{r}}^{x} F(s)\,\mathrm{d}s.}$$

$W(x)$ is the work done *by the force* in moving from that reference position to x.

It is a function of position. It is *not* the extra work the force performs during each subsequent second of motion.

For a constant force F and reference $x_\mathrm{r} = 0$, we get $W(x) = Fx$. If we shift position slightly, $\delta W \simeq F\,\delta x$.

5. Work done by the force versus potential energy

Potential energy uses the opposite sign. For a conservative force,

$$F(x) = -\frac{\mathrm{d}V}{\mathrm{d}x}, \qquad \text{whereas} \qquad F(x) = \frac{\mathrm{d}W}{\mathrm{d}x}.$$

Therefore,

$$\boxed{W(x) = -V(x) + C.}$$

If we choose matching reference constants, we may write $W = -V$.

	$W(x)$	$V(x)$
Meaning	Work done *by* the force from a reference point	Potential energy at the position
Moving with the force	Increases	Decreases
Moving against the force	Decreases	Increases
At stable equilibrium	Local maximum	Local minimum

For example, near Earth's surface with height y measured upwards,

$$V(y) = mgy, \qquad W(y) = -mgy$$

when both are zero at $y = 0$.

Raising the particle requires positive work *against* gravity. Gravity itself does negative work, so W decreases while V increases.

> **Keep this straight.** W means work done *by the force*. Work done *against the force* is $-W$ (for the same reference displacement). Potential energy is the latter, up to an arbitrary constant.

6. Now let the particle move

Return to the journey from A to B in time τ. This time a force acts.

Our free-particle observation still matters: making a velocity profile more uneven tends to increase $\int T\,\mathrm{d}t$ relative to a uniform profile. But a force causes acceleration, so the actual motion generally *cannot* be the uniform-velocity journey.

Something else must enter the description.

Imagine a proposed position–time curve. Alter a small part of it very slightly, while keeping the two endpoints fixed. The altered curve passes through slightly different positions, and its velocity must change a little too.

So one small change to the journey has *two* consequences.

7. Two effects of one small perturbation



Effect one: the motion changes. The perturbation changes the velocities around the altered section. Hence it changes the accumulated kinetic energy $\int T \, \mathrm{d}t$.

Effect two: the force contribution changes. The perturbation changes the positions visited in the force field. At each time, the work function changes by approximately

$$\delta W = F(x) \, \delta x.$$

Hence the accumulated quantity $\int W(x(t)) \, \mathrm{d}t$ changes as well.

The changes can have opposite signs. A perturbation might reduce the kinetic contribution but increase the work contribution, or vice versa.

The central proposal

Perhaps the physical journey is special because these two changes *balance*.

This motivates introducing a quantity associated with an entire journey:

$$\boxed{S[x] = \int_0^\tau (T + W(x)) \, \mathrm{d}t.}$$

We call it the *action*.

> **The intuitive picture.** A free particle avoids unnecessary velocity fluctuations. A force adds a position-dependent work contribution. In actual motion, the first-order changes in these two contributions balance when we slightly alter the whole journey.

8. What exactly is being balanced?

Not T and W themselves. They need not be equal, and neither must be small.

It is their *changes under a small alteration of the path* that balance:

$$\boxed{\delta \int_0^\tau T \, \mathrm{d}t + \delta \int_0^\tau W \, \mathrm{d}t = 0.}$$

The endpoints and elapsed time remain fixed. The balance must hold for *every* sufficiently small permitted perturbation.

9. From small time steps to the full action

Think of the journey as divided into many short intervals of duration Δt.

To each interval, associate a small action contribution,

$$\Delta S \approx (T + W)\Delta t.$$

Add these contributions and make the intervals finer:

$$S = \lim_{\Delta t \to 0} \sum_i (T_i + W_i)\Delta t = \int_0^\tau (T + W)\, \mathrm{d}t.$$

The action has units of energy multiplied by time. It is *not* the particle's total energy, and it is not the total work done along its actual path.

There is also an important local subtlety. A tiny displacement at one instant changes velocities just before and just after that instant. We cannot demand $\delta T + \delta W = 0$ separately at every isolated instant. Instead, the *integrated first-order change* must vanish for arbitrary small perturbations.

10. Why the familiar minus sign appears

We have already learned that, for a conservative force and consistent reference constants,

$$W = -V.$$

So our proposed action becomes

$$\boxed{S[x] = \int_0^\tau (T - V)\, \mathrm{d}t.}$$

The quantity inside the integral is the *Lagrangian*:

$$\boxed{L = T - V.}$$

The minus sign is not telling us that nature "likes" high potential energy. It is a consequence of using potential energy, which is the negative of the work done *by* a conservative force.

What does stationary mean?

The physical path makes the *first-order* change in S vanish:

$$\boxed{\delta S = 0.}$$

For a free particle, this stationary value is a minimum. In general, however, *stationary* does not always mean *smallest*. That is why we say the *principle of stationary action*, not universally the principle of least action.

A helpful distinction. The two simple limiting cases—steady free motion and equilibrium at rest—*motivate* the balance $T + W$. They do not logically force that particular formula. Stationary action is a physical principle whose correctness must also be checked against observed motion.

11. Does the balance give the right motion?

We need only a short check, not a course in variational calculus.

Suppose the force is constant and acts to the right. Then $W(x) = Fx$ and

$$S = \int_0^\tau \left(\frac{1}{2}m\dot{x}^2 + Fx\right) \mathrm{d}t.$$

Imagine changing the path by a tiny amount $\delta x(t)$ that is zero at the endpoints.

The change in the work contribution is

$$\delta \int_0^\tau W \,\mathrm{d}t = \int_0^\tau F\,\delta x \,\mathrm{d}t.$$

The change in the kinetic contribution is initially

$$\delta \int_0^\tau T \,\mathrm{d}t = \int_0^\tau m\dot{x}\,\delta\dot{x} \,\mathrm{d}t.$$

An integration by parts, using the fixed endpoints, rewrites this as

$$\delta \int_0^\tau T \,\mathrm{d}t = -\int_0^\tau m\ddot{x}\,\delta x \,\mathrm{d}t.$$

Therefore the total change is

$$\delta S = \int_0^\tau (F - m\ddot{x})\delta x \,\mathrm{d}t.$$

If this must vanish for *every* small alteration $\delta x(t)$, then

$$\boxed{m\ddot{x} = F.}$$

That is Newton's second law.

The important insight is not the algebra. It is the origin of the two terms:

- The kinetic contribution responds to how the velocity changes and gives the acceleration term.
- The work contribution responds to displacement in the force field and gives the force term.
- Stationarity requires the two effects to balance.

12. Why call this a new formalism?

Newton's description begins with a force and asks for the resulting acceleration. Lagrange's description begins with a single quantity L, assigns an action to each possible history, and identifies the actual history through stationarity.

For this simple particle, both descriptions give the same motion. But the Lagrangian viewpoint often remains useful when coordinates, constraints, and systems become more complicated.

You do not need those applications yet to appreciate the central idea.

13. The whole intuition, in one place

Start with inertia. For a free particle travelling between fixed endpoints in a fixed time, uniform velocity minimizes the accumulated kinetic energy. Fluctuations add a positive contribution because velocity is squared.

Then consider equilibrium. A particle at rest can remain at a position where a tiny displacement produces no first-order work. Stable equilibrium adds the requirement that displacement is opposed by a restoring force.

Now combine the ideas. For a moving particle in a conservative force field, a small change of trajectory changes both the accumulated kinetic energy and the position-dependent work contribution. The actual trajectory is where their first-order changes balance.

$$\delta \int_0^\tau (T + W)\,\mathrm{d}t = 0$$

and, because $W = -V$ with a consistent choice of zero,

$$\boxed{\delta \int_0^\tau (T - V)\,\mathrm{d}t = 0.}$$

What we should remember. The action is not an ordinary cost that nature must always minimize. Its stationarity is a compact way of expressing a balance between the inertial and force-related effects of perturbing a possible history.

This is a motivation, not a proof from first principles that nature *must* choose this law. But it gives the law an intelligible meaning: rather than an unexplained $T - V$, we see two contributions whose changes must fit together.

End of the conceptual introduction